Haagerup Conformal Field Theories from the E_8 Lattice

Tzu-Chen Huang[1]
[1] *[Affiliation]*
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We construct an explicit rational, unitary chiral algebra U at central charge $c = 8$ whose representation category is the Drinfeld center of the Haagerup fusion category $\mathcal{H}_3$. Inside the lattice vertex algebra of $E_6 \oplus A_2$, U is generated by the currents of $\mathfrak{su}(2)_1 \oplus \mathfrak{su}(2)_{39}$, the stress tensor, and a single weight-two field made of three vertex operators with relative phase $(\sqrt{13}+3i\sqrt{3})/\sqrt{40}$. Its diagonal modular invariant is a $c = 8$ conformal field theory with the $Z(\mathcal{H}_3)$ Verlinde lines. The chiral algebra of the $(E_8)_1$ Wess–Zumino–Witten model is a holomorphic extension of U, and the model carries topological lines $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$, as recently proposed. We present the spectra, exact chiral correlators, the action of the Haagerup lines, defect Hilbert spaces, and boundary states, and compare with modular-bootstrap bounds.

Introduction and summary.—Symmetries of two-dimensional quantum field theories are implemented by topological line operators. Dropping the requirement that these lines be invertible leads to non-invertible symmetries, described by fusion categories [1–4]. Among these the Haagerup fusion category $\mathcal{H}_3$ [5–8] is the simplest example that arises from neither finite groups nor affine Lie algebras [9]. It has six simple lines $1, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho$ with

$$\alpha^3 = 1, \qquad \alpha\rho = \rho\alpha^2, \qquad \rho^2 = 1 + \rho + \alpha\rho + \alpha^2\rho, \quad (1)$$

and $\mathrm{qdim}\,\rho = d \equiv (3+\sqrt{13})/2$.

Whether $\mathcal{H}_3$ is the symmetry of a conformal field theory (CFT) has become a test case for non-invertible symmetries. Anyon chains built from $\mathcal{H}_3$ show numerical evidence for a critical point at $c \approx 2$ [10–14]. Topological field theories [15, 16], a gapless phase with degenerate vacua [17], and non-unitary rational candidates [18] with Haagerup symmetry have also been found. Earlier, building on Izumi's computation of the modular data of the Drinfeld center $Z(\mathcal{H}_3)$ [7], Evans and Gannon [19] found candidate characters of a rational CFT at $c = 8$ with these modular data, related to $(E_8)_1$ and $(E_6)_1 \times (A_2)_1$. Recently, Albert, Honda, Kaidi, and Zheng [20] proposed that the $(E_8)_1$ Wess–Zumino–Witten (WZW) model has $\mathcal{H}_3 \times \mathcal{H}_3^{\mathrm{op}}$ symmetry, and obtained it by gauging such a $Z(\mathcal{H}_3)$ theory, whose existence they assumed. Gannon and Rayhaun [21] showed that its chiral algebra would be the fixed-point algebra of a hypergroup action on $(E_6)_1 \otimes (A_2)_1$.

In this Letter we construct this theory explicitly. The main results are as follows. (i) We give a chiral algebra $U \subset (E_6)_1 \otimes (A_2)_1$ at $c = 8$, generated by the currents of $\mathfrak{su}(2)_1 \oplus \mathfrak{su}(2)_{39}$, the stress tensor, and one explicit weight-two field u, Eq. (4). (ii) U is rational and unitary with $\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3)$, and its twelve characters are those of [19]. (iii) Two full CFTs follow: the diagonal theory of U, whose Verlinde lines form $Z(\mathcal{H}_3)$, and the $(E_8)_1$ WZW model, whose lines commuting with $U \otimes \overline{U}$ form $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$. The latter establishes the Haagerup symmetry of $(E_8)_1$ proposed in [20]. (iv) Because both theories are rational, their data are exactly computable. We give spectra, correlators, operator product expansion (OPE) data, twisted Hilbert spaces, and boundary states.

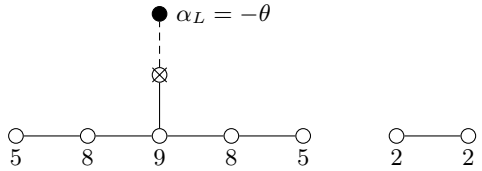


FIG. 1. Dynkin diagram of $E_6 \oplus A_2$, with the E_6 part extended by its lowest root. The simple roots $\alpha_0, \ldots, \alpha_4$ run left to right along the bottom row of E_6, α_5 is the crossed node, and α_6, α_7 span A_2. The black node $\alpha_L = -\theta$ generates $\mathfrak{su}(2)_1$. The numbers are the coefficients c_i of the principal $\mathfrak{su}(2)$ of $A_5 \oplus A_2$, which generates $\mathfrak{su}(2)_{39}$. The three momenta in u are $a_1 = \alpha_6 + \alpha_7 - \alpha_5$, $a_2 = \alpha_3 + \alpha_4 - \alpha_5$, and $a_3 = \alpha_0 + \alpha_1 - \alpha_5$.

The existence question reduces to a finite set of exact algebraic identities in a lattice vertex operator algebra (VOA), together with further exact checks on explicit fields, all of which we have verified by computer algebra. The representation theory then follows from structural arguments. The complete mathematical treatment will appear in a companion paper [22]. Here we concentrate on the construction and on the physics.

The chiral algebra.—We start from the lattice VOA $B = V_{E_6 \oplus A_2}$, the chiral algebra of the $(E_6)_1 \times (A_2)_1$ WZW model, realized by eight chiral bosons $X(z)$ on the $E_6 \oplus A_2$ root lattice [23, 24]. Label the simple roots as in Fig. 1, and let $\alpha_L = -\theta$ be the lowest root of E_6. It is orthogonal to the A_5 subdiagram $\alpha_0, \ldots, \alpha_4$. The currents

$$H_{(1)} = \alpha_L \cdot i\partial X, \qquad E_{(1)} = e^{i\alpha_L \cdot X} \quad (2)$$

and their conjugate generate $\mathfrak{su}(2)_1$. The principal $\mathfrak{su}(2)$ subalgebra of $\mathfrak{su}(6) \oplus \mathfrak{su}(3)$ has

$$H_{(39)} = \sum_i c_i\, \alpha_i \cdot i\partial X, \qquad E_{(39)} = \sum_i \sqrt{c_i}\, e^{i\alpha_i \cdot X}, \quad (3)$$

with $(c_0, \ldots, c_4; c_6, c_7) = (5, 8, 9, 8, 5; 2, 2)$ and $c_5 = 0$. Its level is $35 + 4 = 39$, the sum of the two embedding

indices. The two $\mathfrak{su}(2)$ algebras commute. We define

$$u = e^{ia_1\cdot X} + \zeta\, e^{ia_2\cdot X} + \bar{\zeta}\, e^{ia_3\cdot X}, \qquad \zeta = \frac{\sqrt{13}+3i\sqrt{3}}{\sqrt{40}}, \tag{4}$$

with the momenta a_i of Fig. 1 [25]. All three have $a_i^2 = 4$, and $a_i \cdot a_j = 2$ for $i \neq j$. Note that $|\zeta| = 1$. The field u is a Virasoro primary of weight two. It is the highest-weight state of spin $(\frac{1}{2}, \frac{13}{2})$ under $\mathfrak{su}(2)_1 \oplus \mathfrak{su}(2)_{39}$, with charges $(1, 13)$ under $(H_{(1)}, H_{(39)})$. Our chiral algebra is the algebra generated by these fields under the OPE,

$$U = \big\langle\, \mathfrak{su}(2)_1 \oplus \mathfrak{su}(2)_{39},\ T,\ u \,\big\rangle \subset B, \tag{5}$$

where T is the stress tensor of B, with $c = 8$.

Because U sits inside a lattice VOA, every OPE follows from free-field Wick contractions, and locality and associativity are automatic. The nontrivial question is whether u generates a proper subalgebra. It could instead regenerate the remaining 80 currents of B. We find that it does not: U is strongly generated by fields of weight at most four, and

$$\begin{aligned} U_1 &= \mathfrak{su}(2) \oplus \mathfrak{su}(2), \\ \chi_U &= q^{-1/3}\big(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots\big). \end{aligned} \tag{6}$$

The stress tensor decomposes into four commuting Virasoro fields,

$$T = T^{\mathfrak{su}(2)_1} + T^{\mathfrak{su}(2)_{39}} + T_e + T_f, \quad 8 = 1 + \tfrac{117}{41} + \tfrac{2400}{1271} + \tfrac{70}{31}. \tag{7}$$

The 120 weight-two fields comprise 22 affine vacuum descendants (16 current bilinears and 6 current derivatives), T_e, T_f, and eight multiplets of $\mathfrak{su}(2)_1 \oplus \mathfrak{su}(2)_{39}$ with dimensions $(1,5)$, $(1,9)^2$, $(1,13)$, $(2,2)$, $(2,6)$, $(2,8)$, and $(2,14)$. The last multiplet contains u. U also contains the W-algebras of the three-state Potts model ($c = \frac{4}{5}$) [26] and of the $\mathbb{Z}_{13}$ parafermions ($c = \frac{8}{5}$) [27]. These echo the numbers 3 and 13 in d.

We found u by requiring that current descendants of a few highest-weight fields span the 120-dimensional weight-two space predicted by the character, and that their singular OPEs close without producing the other 80 currents of B. The phase ζ is the resulting solution within one ansatz for these fields and one of two possible current embeddings; we have not established its uniqueness. Closure in all weights follows from a finite strong-generation criterion. It reduces to a finite list of OPE identities, which we verified exactly over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$ [22]. In particular $U \neq B$, since $\dim B_1 = 86$. U is stable under the PCT operation of B and inherits its positive inner product, so it is unitary. Its stress tensor is a positive combination of stress tensors of rational subalgebras: the two affine algebras and conjugates of the Potts and parafermion algebras. This implies that U is C_2-cofinite and rational. Combining this with the branching of lattice modules, fusion-ring and modular-data constraints, the classification of unitary Haagerup-type categories [7, 8], and an analytic argument for the unitarity of all modules, we find

TABLE I. The twelve irreducible U modules. Labels follow [20], with those of [19] in the second column. Here h is the conformal weight, t_M the dimension of the lowest-weight space, and $X = \rho \oplus \alpha\rho \oplus \alpha^2\rho$. The last column is the restriction $F(M)$ to $\mathcal{H}_3$, which gives the Haagerup content of the E_8 theory. All sectors are self-conjugate, with twists $e^{2\pi i h}$.

M	[19]	h	t_M	qdim M	$F(M)$
1	0	0	1	1	1
π_1	b	1	80	d^2	$1 \oplus X$
π_2	a	1	81	$2+3d$	$2\cdot 1 \oplus X$
σ_1	c_0	1	81	$2+3d$	$\alpha \oplus \alpha^2 \oplus X$
σ_2	c_1	$\frac{1}{3}$	3	$2+3d$	$\alpha \oplus \alpha^2 \oplus X$
σ_3	c_2	$\frac{2}{3}$	27	$2+3d$	$\alpha \oplus \alpha^2 \oplus X$
μ_1	d_3	$\frac{15}{13}$	119	$3d$	X
μ_2	d_2	$\frac{11}{13}$	42	$3d$	X
μ_3	d_4	$\frac{5}{13}$	5	$3d$	X
μ_4	d_6	$\frac{8}{13}$	14	$3d$	X
μ_5	d_1	$\frac{6}{13}$	7	$3d$	X
μ_6	d_5	$\frac{7}{13}$	13	$3d$	X

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3) \tag{8}$$

as unitary modular tensor categories [22]. The characters of U are the $\gamma = 0$ solution of [19] [cf. Eq. (12) of [20]].

The twelve modules are listed in Table I. The modular S matrix is that of $Z(\mathcal{H}_3)$ [19]. The total quantum dimension is $\mathcal{D} = 3(2+3d) = 3+3d^2$, and $\mathcal{D}^2$ is the global dimension. As U modules, $(E_6)_1 \otimes (A_2)_1$ and, for a suitable embedding $U \subset V_{E_8}$ [28], $(E_8)_1$ decompose as

$$(E_6)_1 \otimes (A_2)_1 = M_1 \oplus M_{\pi_1}, \qquad (E_8)_1 = M_1 \oplus M_{\pi_1} \oplus 2M_{\pi_2}. \tag{9}$$

For example, the 248 currents of E_8 split as $6+80+81+81$. These are the chiral counterparts of the gaugings in [20]: they correspond to the commutative algebras $1 \oplus \pi_1$ and $1 \oplus \pi_1 \oplus 2\pi_2$ of $Z(\mathcal{H}_3)$ [28]. The first exhibits U as the fixed-point algebra anticipated in [21].

Two full CFTs.—The diagonal modular invariant

$$Z_{\mathrm{diag}} = \sum_M |\chi_M(\tau)|^2 \tag{10}$$

defines a unitary CFT with $c_L = c_R = 8$ and twelve $U \otimes \overline{U}$ primaries. We call it the $Z(\mathcal{H}_3)$ theory. Its topological lines commuting with $U \otimes \overline{U}$ are the twelve Verlinde lines [29, 30]. Their fusion rules are those of $Z(\mathcal{H}_3)$, and L_a acts on sector $M_i \otimes \overline{M_i}$ by S_{ai}/S_{1i}.

The second full CFT is the $(E_8)_1$ WZW model, $Z_{E_8} = |\chi_{E_8}|^2$ with $\chi_{E_8} = E_4/\eta^8$. By Eq. (9), it is the extension of $U \otimes \overline{U}$ by the Lagrangian algebra $A = 1 \oplus \pi_1 \oplus 2\pi_2$ in each chirality. Its topological lines commuting with $U \otimes \overline{U}$

therefore form $\mathrm{Rep}(U)_A \boxtimes \mathrm{Rep}(U)_A^{\mathrm{op}}$, where $\mathrm{Rep}(U)_A$ is the category of all (not necessarily local) A-modules [1, 31]. For this A, $\mathrm{Rep}(U)_A \simeq \mathcal{H}_3$. The $(E_8)_1$ WZW model therefore has $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ symmetry, as proposed in [20]. These lines commute with U but not with the full E_8 current algebra. Gauging the diagonal $\mathcal{H}_3$ returns the $Z(\mathcal{H}_3)$ theory [20].

Below we mostly use the first factor $\mathcal{H}_3 \boxtimes 1$, which acts on the left movers only. The untwisted Hilbert space is $(M_1 \oplus M_{\pi_1} \oplus \mathbb{C}^2 \otimes M_{\pi_2}) \otimes \overline{V}_{E_8}$. In a suitable basis the lines act as

$$\widehat{\alpha} = \big(1,\, 1,\, \mathrm{diag}(\omega, \omega^2)\big), \qquad \widehat{\rho} = \big(d,\, -d^{-1},\, \sigma_x\big), \quad (11)$$

with $\omega = e^{2\pi i/3}$. They satisfy Eq. (1). In particular, the 248 currents split into 6 that are invariant ($\widehat{\rho} = d = \langle\rho\rangle$), 80 with $\widehat{\alpha} = 1$ and $\widehat{\rho} = -d^{-1}$, and 2×81 with $\widehat{\alpha} = \omega^{\pm 1}$ that $\widehat{\rho}$ exchanges. The α-invariant subalgebra $U \oplus M_{\pi_1}$ has the character of $(E_6)_1 \otimes (A_2)_1$, consistent with identifying α with the $\mathbb{Z}_3$ automorphism of E_8 of class 3B [20]. On the $\mathbb{Z}_3$-neutral states $\widehat{\rho}$ takes the two values $(3 \pm \sqrt{13})/2$, just as in the lattice model at $c \approx 2$ [11]; see Fig. 2(b). The twined partition functions are

$$Z^{[\alpha]} = (\chi_1 + \chi_{\pi_1} - \chi_{\pi_2})\overline{\chi}_{E_8}, \quad Z^{[\rho]} = (d\chi_1 - d^{-1}\chi_{\pi_1})\overline{\chi}_{E_8}. \quad (12)$$

The defect Hilbert spaces follow from the restriction F in Table I:

$$Z_\alpha = \Big(\sum_{i=1}^{3} \chi_{\sigma_i}\Big)\overline{\chi}_{E_8}, \quad Z_\rho = \Big(\sum_{M \neq 1} \chi_M\Big)\overline{\chi}_{E_8}. \quad (13)$$

These are chiral versions of Eqs. (15)–(16) of [20]. Spins in the α-twisted sector lie in $\frac{1}{3}\mathbb{Z}$, as expected for a non-anomalous $\mathbb{Z}_3$. The ρ-twisted sector contains the additional spins $6j^2/13 \bmod 1$, i.e. $\{\frac{2}{13}, \frac{5}{13}, \frac{6}{13}, \frac{7}{13}, \frac{8}{13}, \frac{11}{13}\}$. In both sectors the lightest defect operator has $(h, \bar{h}) = (\frac{1}{3}, 0)$, with multiplicity three.

Spectrum.—Because all conformal weights are known, the spectrum follows from the characters. Removing Virasoro descendants is straightforward at $c = 8 > 1$, where Verma modules of positive weight are irreducible. Table II and Fig. 2 summarize the two theories. The diagonal theory has seven relevant scalar levels,

Δ	$\frac{2}{3}$	$\frac{10}{13}$	$\frac{12}{13}$	$\frac{14}{13}$	$\frac{16}{13}$	$\frac{4}{3}$	$\frac{22}{13}$
mult.	9	25	49	169	196	729	1764

(14)

coming from $\sigma_2, \mu_3, \mu_5, \mu_6, \mu_4, \sigma_3, \mu_2$. Each multiplicity is t_M^2. The lightest scalar, at $\Delta = \frac{2}{3}$, transforms as $(\mathbf{1}, \mathbf{3}) \otimes \overline{(\mathbf{1}, \mathbf{3})}$ under the currents. Including spin, there are 10623 relevant Virasoro primaries. At $\Delta = 2$ there are $80^2 + 81^2 + 81^2 + 6^2 = 19558$ scalar primaries. The E_8 theory has scalar gap 2 and no relevant scalar.

Neither theory has a symmetric relevant deformation. The E_8 theory has no relevant scalar at all. In the diagonal theory, a sector fixed by all Verlinde lines has $S_{ai}/S_{1i} = \mathrm{qdim}\, L_a$ for every a, which forces $i = 1$ because S is invertible. The symmetric operators are therefore those of $U \otimes \overline{U}$, and the lightest symmetric scalars are the 36 current bilinears $J^a \bar{J}^b$ at $\Delta = 2$. All 2941 relevant scalars are charged, so the $Z(\mathcal{H}_3)$ symmetry alone keeps the diagonal theory critical without tuning. In the E_8 theory the $\mathcal{H}_3$-invariant subspace is $U \otimes \overline{V}_{E_8}$, with lightest scalars $J_U^a \bar{J}_{E_8}^B$.

Correlators and OPE data.—The lattice realization gives explicit interacting correlators. Let $\hat{u} = u/\sqrt{3}$ be the unit normalized field and $\hat{u}^\dagger$ its conjugate. Its four-point functions are rational:

$$\langle \hat{u}^\dagger(\infty)\hat{u}(1)\hat{u}^\dagger(x)\hat{u}(0)\rangle = \frac{3 - 4x + 4x^2}{3x^4(1-x)^4}, \quad (15)$$

$$\langle \hat{u}^\dagger(\infty)\hat{u}^\dagger(1)\hat{u}(x)\hat{u}(0)\rangle = \frac{x^2(4 - 4x + 3x^2)}{3(1-x)^4}. \quad (16)$$

In the neutral channel, write $x^4 G = A_0 + \sum_{n \geq 1} A_n x^n {}_2F_1(n, n; 2n; x)$. Then $(A_0, \ldots, A_4) = (1, \frac{8}{3}, \frac{14}{3}, \frac{58}{9}, \frac{107}{15})$. Here $A_1 = \frac{1}{2} + \frac{13}{6}$ is the exchange of the two Cartan currents. The stress tensor contributes $2h^2/c = 1$ to A_2, and the remaining $\frac{11}{3}$ is the sum of squared OPE coefficients of $\hat{u} \times \hat{u}^\dagger$ into weight-two Virasoro primaries. In the equal-charge channel, the OPE $\hat{u} \times \hat{u}$ starts at x^2 with a single weight-six primary of squared coefficient $\frac{4}{3}$.

Beyond these, we computed every three-point coefficient among the 113 weight-two Virasoro primaries of U. Of these 18365 are nonzero; all are exact, with squared magnitudes in $\mathbb{Q}(\sqrt{13})$. We also computed the four-point functions of the spin-three generators of the embedded Potts and parafermion algebras. Together with spectra through $\Delta = 24$ and the defect and boundary data below, these are provided as ancillary files.

Boundaries.—The diagonal theory has twelve Cardy states [32], with g-values $g_a = \mathrm{qdim}\, M_a/\sqrt{\mathcal{D}}$. The E_8 theory has six elementary U-preserving boundaries, labeled by $x \in \mathcal{H}_3$. Their annulus partition functions are $Z_{xy} = \sum_{z,M} N_{x^* y}^z\, n_{zM}\, \chi_M$, where $n_{zM} = \dim \mathrm{Hom}(F(M), z)$. Their g-values are 1 for x invertible

TABLE II. The two full CFTs, with $c_L = c_R = 8$. Degeneracies count Virasoro primaries. "Symmetric" means invariant under all Verlinde lines (diagonal theory) or under the first-factor $\mathcal{H}_3$ (E_8 theory).

	$Z(\mathcal{H}_3)$ theory	$(E_8)_1$ WZW
Currents per chirality	6	248
Lightest primary Δ	$\frac{2}{3}$	1
Lightest scalar Δ	$\frac{2}{3}$	2
Relevant scalars	2941	0
Scalars at $\Delta = 2$	19558	61504
Symmetric scalar gap (mult.)	2 (36)	2 (1488)
Lines commuting with $U \otimes \overline{U}$	$Z(\mathcal{H}_3)$	$\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$

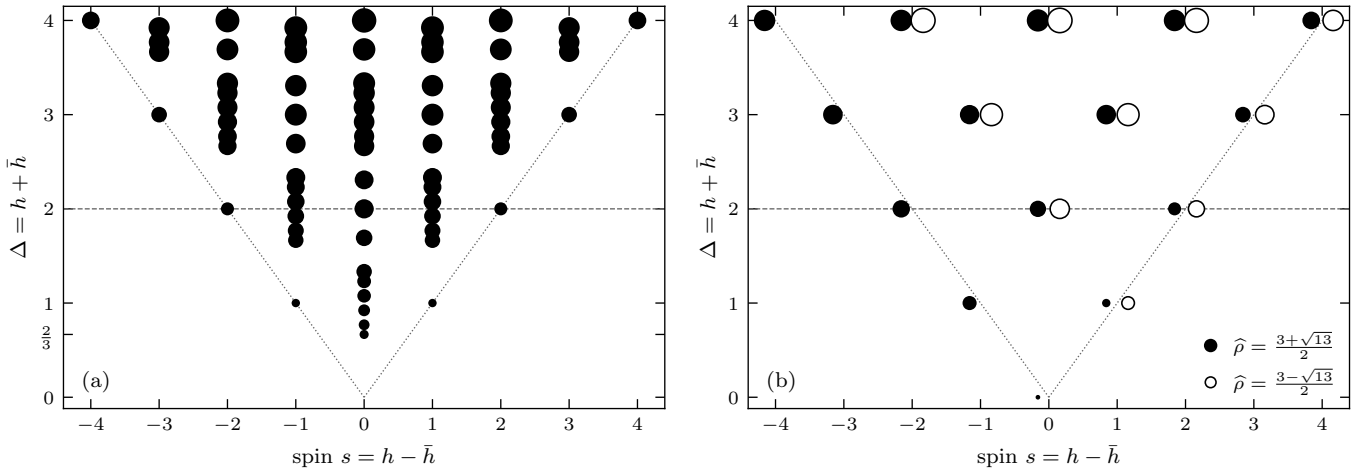


FIG. 2. Exact Virasoro-primary spectra; marker area grows with the logarithm of the multiplicity. (a) The diagonal $Z(\mathcal{H}_3)$ theory for $0 < \Delta \leq 4$. The seven relevant scalar levels are listed in Eq. (14). (b) The $(E_8)_1$ WZW model, showing the $\mathbb{Z}_3$-neutral states of the first-factor $\mathcal{H}_3$ action with $\Delta \leq 4$, including the identity. Filled markers: $\widehat{\rho} = (3+\sqrt{13})/2$, i.e. $M_1 \otimes \overline{V}_{E_8}$. Hollow markers: $\widehat{\rho} = (3-\sqrt{13})/2$, i.e. $M_{\pi_1} \otimes \overline{V}_{E_8}$; where both occur, they are offset horizontally. Compare Fig. 6 of [11]. Dashed lines: $\Delta = 2$. Dotted lines: $\Delta = |s|$.

and d for $x = \alpha^k \rho$. The invertible boundaries are the ordinary E_8 boundary and its $\mathbb{Z}_3$ images. Each ρ-type boundary supports nine relevant boundary operators of weight $\frac{1}{3}$.

Bootstrap.—In the ordinary modular bootstrap [33], the E_8 theory sits at the known kink $(c, \Delta_{\text{scalar}}) = (8, 2)$, and the diagonal theory at $(8, \frac{2}{3})$. For the Haagerup-symmetric bootstrap of [20], based on the methods of [34, 35], the relevant sector functions are $\chi_M \overline{\chi}_{E_8}$ [36]. They give an $\mathcal{H}_3$-invariant scalar gap $\Delta = 2$, and a scalar gap $\Delta = 2$ in the $\mathbb{Z}_3$-charged tube sectors $\pi_2 \cup \sigma_1$. Both points lie inside the allowed regions of Figs. 2 and 3 of [20], below the numerical bounds at $c = 8$. The bounds are therefore not saturated by these theories.

Discussion and outlook.—We have given an explicit, rational, and unitary CFT whose topological lines realize the Haagerup category $\mathcal{H}_3$ and its center. The construction is elementary: eight free bosons, the currents of Eqs. (2)–(3), and the field u of Eq. (4).

Conformal manifolds with Haagerup symmetry. The lines of (11) commute with the currents of U and with all antiholomorphic E_8 currents. Deformations by $J_U^a \bar{J}^b$, with J_U^a the two Cartan currents of U and $\bar{J}^b$ the eight Cartan currents of $\overline{E}_8$, are exactly marginal [37]. They span a family of Narain CFTs locally isomorphic to $O(2,8)/O(2) \times O(8)$, before global identifications. Because the perturbing operator passes through every first-factor line, standard arguments [4] suggest that the $\mathcal{H}_3$ symmetry survives on this sixteen-dimensional family, most of which is irrational. The second factor $\mathcal{H}_3^{\text{op}}$ is generically broken. Likewise, the $Z(\mathcal{H}_3)$ Verlinde lines of the diagonal theory should survive its four $J_U \bar{J}_U$ deformations. Establishing this would give continuous families of CFTs with a unique vacuum and Haagerup (or $Z(\mathcal{H}_3)$) symmetry, in the spirit of the non-invertible symmetries found along the $c = 1$ moduli space [38].

Bulk OPE and crossing. The OPE of the nonvacuum sectors of the diagonal theory is not yet computed, beginning with the nine $\Delta = \frac{2}{3}$ scalars. Lattice intertwiners give explicit chiral blocks. For instance the lightest module has blocks $x^{-2/3}(1-x)^{-\beta_i \cdot \beta_j}$, where β_i are the A_2 weights. Sewing these into single-valued correlators would provide a fully solvable benchmark for bootstraps with non-invertible symmetry [35].

Relation to $c \approx 2$. Our theories are not the continuum limit of the $\mathcal{H}_3$ anyon chains [11, 12]. In [20] it was proposed that asymmetric quotients related to $(A_2)_1$ and $(E_6)_1$ yield Haagerup-symmetric theories at $c = 2$ and 6, where the bootstrap bounds have kinks. Our explicit chiral algebra may help to test this.

Other exotic categories. Gannon, Schopieray, and Yadav have now constructed Haagerup–Izumi categories for every odd cyclic group and proven the Evans–Gannon form of their centers [39]. A natural next step is to realize these centers by chiral subalgebras of lattice VOAs. Our method provides a template: guess generators from the predicted character, then certify closure with a finite strong-generation criterion. It would also be interesting to find Haagerup-symmetric lattice models whose continuum limit is the $c = 8$ theory, and to study the relevant boundary perturbations above.

Exact characters, spectra, OPE tensors, and defect and boundary data accompany this preprint.

[To be added.]

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